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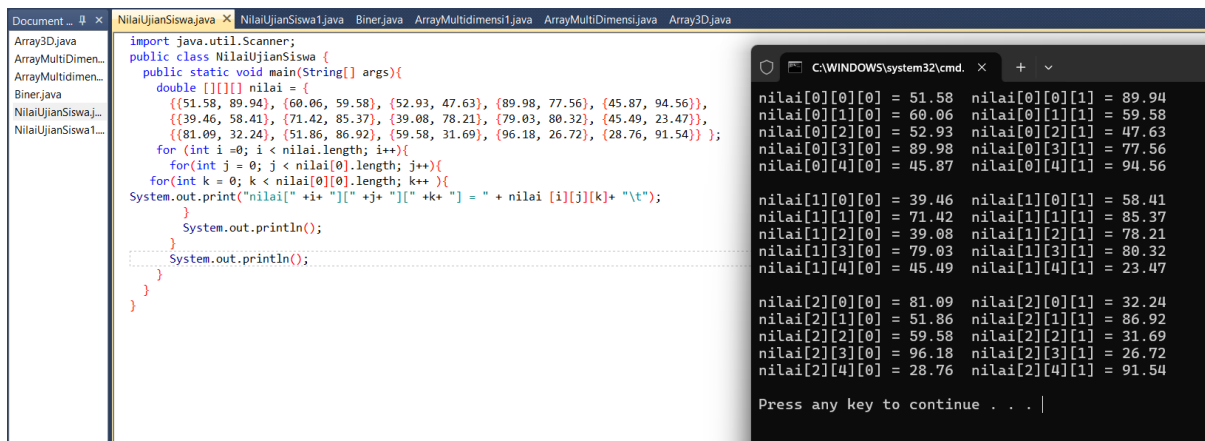
LISTING PRAKTIKUM ALGORITMA DAN PEMROGRAMAN LANJUT

MODUL 5

ARRAY 3 DIMENSI

PRAKTIK

Praktik 1



```
import java.util.Scanner;
public class NilaiUjianSiswa {
    public static void main(String[] args){
        double [][][] nilai = {
            {{51.58, 89.94}, {60.06, 59.58}, {52.93, 47.63}, {89.98, 77.56}, {45.87, 94.56}},
            {{39.46, 58.41}, {71.42, 85.37}, {39.08, 78.21}, {79.03, 80.32}, {45.49, 23.47}},
            {{81.09, 32.24}, {51.86, 86.92}, {59.58, 31.69}, {96.18, 26.72}, {28.76, 91.54}}
        };
        for (int i = 0; i < nilai.length; i++){
            for (int j = 0; j < nilai[i].length; j++){
                for (int k = 0; k < nilai[i][j].length; k++){
                    System.out.print("nilai[" + i + "][" + j + "][" + k + "] = " + nilai[i][j][k] + "\t");
                }
                System.out.println();
            }
        }
    }
}
```

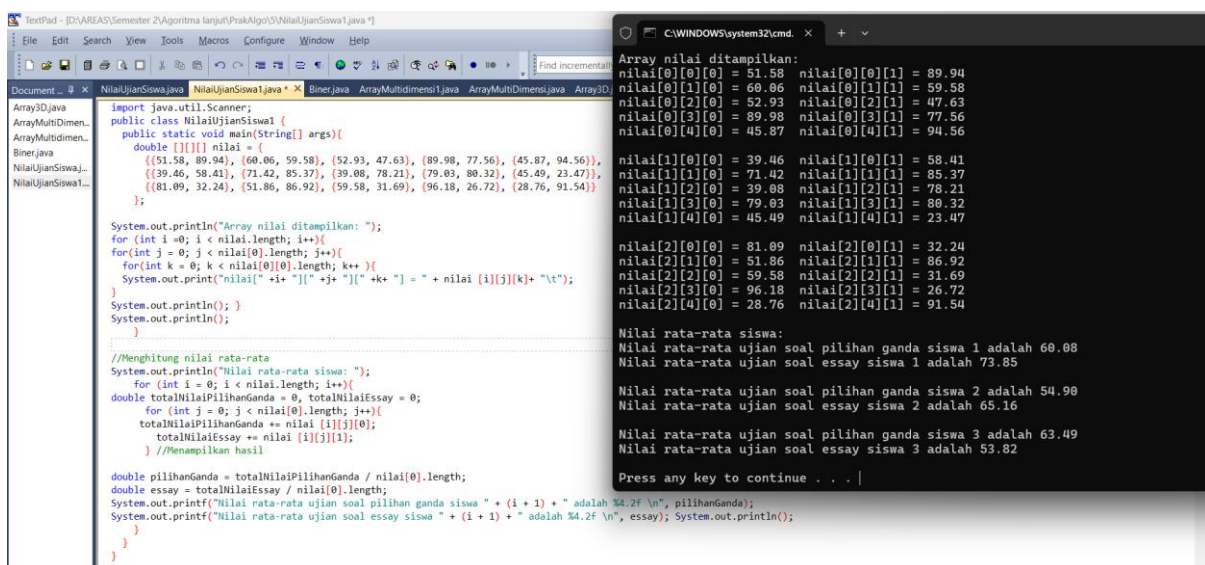
```
nilai[0][0][0] = 51.58 nilai[0][0][1] = 89.94
nilai[0][1][0] = 60.06 nilai[0][1][1] = 59.58
nilai[0][2][0] = 52.93 nilai[0][2][1] = 47.63
nilai[0][3][0] = 89.98 nilai[0][3][1] = 77.56
nilai[0][4][0] = 45.87 nilai[0][4][1] = 94.56

nilai[1][0][0] = 39.46 nilai[1][0][1] = 58.41
nilai[1][1][0] = 71.42 nilai[1][1][1] = 85.37
nilai[1][2][0] = 39.08 nilai[1][2][1] = 78.21
nilai[1][3][0] = 79.03 nilai[1][3][1] = 80.32
nilai[1][4][0] = 45.49 nilai[1][4][1] = 23.47

nilai[2][0][0] = 81.09 nilai[2][0][1] = 32.24
nilai[2][1][0] = 51.86 nilai[2][1][1] = 86.92
nilai[2][2][0] = 59.58 nilai[2][2][1] = 31.69
nilai[2][3][0] = 96.18 nilai[2][3][1] = 26.72
nilai[2][4][0] = 28.76 nilai[2][4][1] = 91.54

Press any key to continue . . . |
```

Praktik 2



```
import java.util.Scanner;
public class NilaiUjianSiswa {
    public static void main(String[] args){
        double [][][] nilai = {
            {{51.58, 89.94}, {60.06, 59.58}, {52.93, 47.63}, {89.98, 77.56}, {45.87, 94.56}},
            {{39.46, 58.41}, {71.42, 85.37}, {39.08, 78.21}, {79.03, 80.32}, {45.49, 23.47}},
            {{81.09, 32.24}, {51.86, 86.92}, {59.58, 31.69}, {96.18, 26.72}, {28.76, 91.54}}
        };
        System.out.println("Array nilai ditampilkan:");
        for (int i = 0; i < nilai.length; i++){
            for (int j = 0; j < nilai[i].length; j++){
                for (int k = 0; k < nilai[i][j].length; k++){
                    System.out.print("nilai[" + i + "][" + j + "][" + k + "] = " + nilai[i][j][k] + "\t");
                }
                System.out.println();
            }
        }
        //Menghitung nilai rata-rata
        System.out.println("Nilai rata-rata siswa:");
        for (int i = 0; i < nilai.length; i++){
            double totalNilaiPilihanGanda = 0, totalNilaiEssay = 0;
            for (int j = 0; j < nilai[i].length; j++){
                totalNilaiPilihanGanda += nilai[i][j][0];
                totalNilaiEssay += nilai[i][j][1];
            }
            //Menampilkan hasil
            double pilihanGanda = totalNilaiPilihanGanda / nilai[i].length;
            double essay = totalNilaiEssay / nilai[i].length;
            System.out.printf("Nilai rata-rata ujian soal pilihan ganda siswa %d adalah %.2f \n", pilihanGanda, pilihanGanda);
            System.out.printf("Nilai rata-rata ujian soal essay siswa %d adalah %.2f \n", essay, essay);
            System.out.println();
        }
    }
}
```

```
Array nilai ditampilkan:
nilai[0][0][0] = 51.58 nilai[0][0][1] = 89.94
nilai[0][1][0] = 60.06 nilai[0][1][1] = 59.58
nilai[0][2][0] = 52.93 nilai[0][2][1] = 47.63
nilai[0][3][0] = 89.98 nilai[0][3][1] = 77.56
nilai[0][4][0] = 45.87 nilai[0][4][1] = 94.56

nilai[1][0][0] = 39.46 nilai[1][0][1] = 58.41
nilai[1][1][0] = 71.42 nilai[1][1][1] = 85.37
nilai[1][2][0] = 39.08 nilai[1][2][1] = 78.21
nilai[1][3][0] = 79.03 nilai[1][3][1] = 80.32
nilai[1][4][0] = 45.49 nilai[1][4][1] = 23.47

nilai[2][0][0] = 81.09 nilai[2][0][1] = 32.24
nilai[2][1][0] = 51.86 nilai[2][1][1] = 86.92
nilai[2][2][0] = 59.58 nilai[2][2][1] = 31.69
nilai[2][3][0] = 96.18 nilai[2][3][1] = 26.72
nilai[2][4][0] = 28.76 nilai[2][4][1] = 91.54

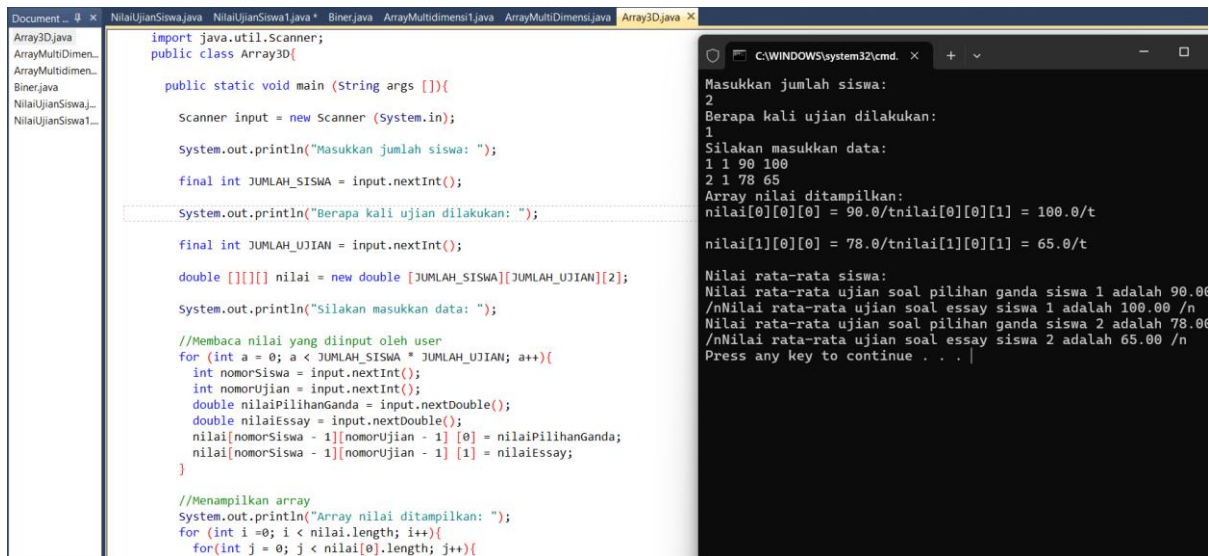
Nilai rata-rata siswa:
Nilai rata-rata ujian soal pilihan ganda siswa 1 adalah 60.08
Nilai rata-rata ujian soal essay siswa 1 adalah 73.85

Nilai rata-rata ujian soal pilihan ganda siswa 2 adalah 54.90
Nilai rata-rata ujian soal essay siswa 2 adalah 65.16

Nilai rata-rata ujian soal pilihan ganda siswa 3 adalah 63.49
Nilai rata-rata ujian soal essay siswa 3 adalah 53.82

Press any key to continue . . . |
```

Praktik 3



The screenshot shows an IDE with a file explorer on the left containing files like `Array3D.java`, `ArrayMultiDimensi1.java`, `ArrayMultiDimensi2.java`, `Array3D.java`, `NilaiUjianSiswa1.java`, `NilaiUjianSiswa2.java`, and `NilaiUjianSiswa3.java`. The main editor displays the code for `Array3D.java`, which includes a `main` method that takes command-line arguments for the number of students and exams, reads input data, and stores it in a 3D array. The code also includes logic to calculate and display the average scores for each student.

```
import java.util.Scanner;
public class Array3D{

    public static void main (String args []){

        Scanner input = new Scanner (System.in);

        System.out.println("Masukkan jumlah siswa: ");

        final int JUMLAH_SISWA = input.nextInt();

        System.out.println("Berapa kali ujian dilakukan: ");

        final int JUMLAH_UJIAN = input.nextInt();

        double [][][] nilai = new double [JUMLAH_SISWA][JUMLAH_UJIAN][2];

        System.out.println("Silakan masukkan data: ");

        //Membaca nilai yang diinput oleh user
        for (int a = 0; a < JUMLAH_SISWA * JUMLAH_UJIAN; a++){
            int nomorSiswa = input.nextInt();
            int nomorUjian = input.nextInt();
            double nilaiPilihanGanda = input.nextDouble();
            double nilaiEssay = input.nextDouble();
            nilai[nomorSiswa - 1][nomorUjian - 1][0] = nilaiPilihanGanda;
            nilai[nomorSiswa - 1][nomorUjian - 1][1] = nilaiEssay;
        }

        //Menampilkan array
        System.out.println("Array nilai ditampilkan: ");
        for (int i = 0; i < nilai.length; i++){
            for (int j = 0; j < nilai[i].length; j++){
```

The command prompt window on the right shows the execution of the program. It prompts for the number of students (2) and the number of exams (1). It then prompts for data entry, where the user enters scores for two students across one exam. The output shows the calculated average scores for each student.

```
Masukkan jumlah siswa:
2
Berapa kali ujian dilakukan:
1
Silakan masukkan data:
1 1 90 100
2 1 78 65
Array nilai ditampilkan:
nilai[0][0][0] = 90.0/tnilai[0][0][1] = 100.0/t
nilai[1][0][0] = 78.0/tnilai[1][0][1] = 65.0/t

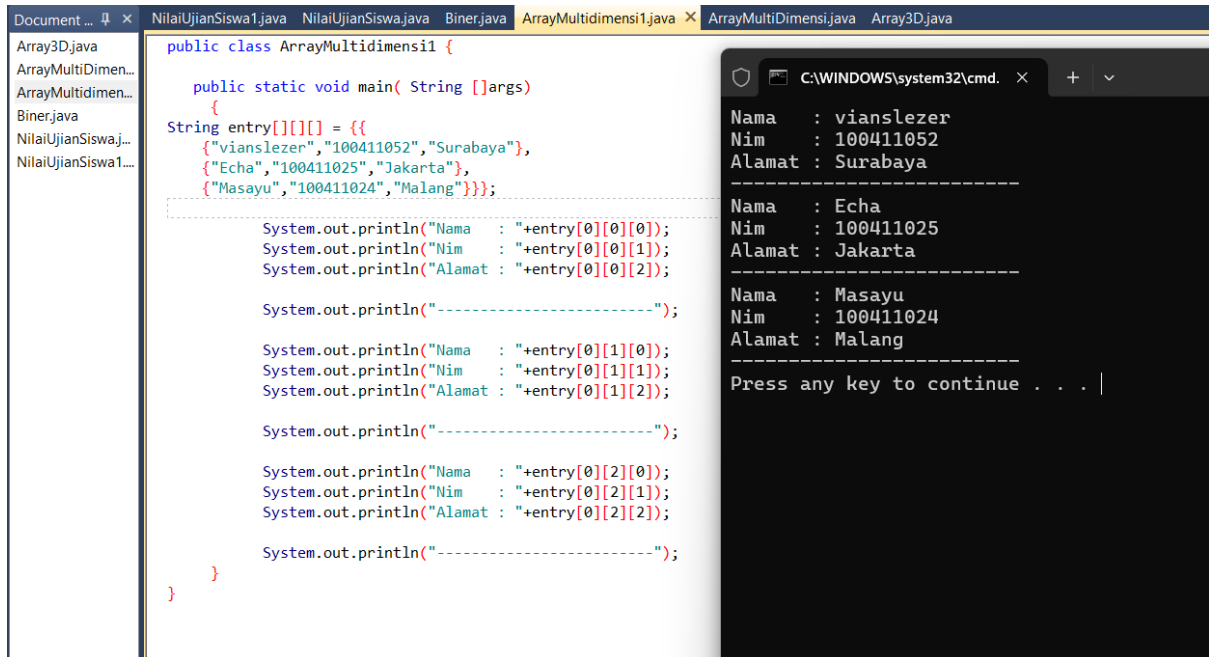
Nilai rata-rata siswa:
Nilai rata-rata ujian soal pilihan ganda siswa 1 adalah 90.00
/nNilai rata-rata ujian soal essay siswa 1 adalah 100.00 /n
Nilai rata-rata ujian soal pilihan ganda siswa 2 adalah 78.00
/nNilai rata-rata ujian soal essay siswa 2 adalah 65.00 /n
Press any key to continue . . . |
```

```
//Menampilkan array
System.out.println("Array nilai ditampilkan: ");
for (int i = 0; i < nilai.length; i++){
    for (int j = 0; j < nilai[i].length; j++){
        for (int k = 0; k < nilai[i][j].length; k++){
            System.out.print("nilai[" + i + "][" + j + "][" + k + "] = " + nilai [i][j][k] + "/t");
        }
        System.out.println();
    }
    System.out.println();
}

//Menghitung nilai rata-rata
System.out.println("Nilai rata-rata siswa: ");
for (int i = 0; i < nilai.length; i++){
    double totalNilaiPilihanGanda = 0, totalNilaiEssay = 0;
    for (int j = 0; j < nilai[i].length; j++){
        totalNilaiPilihanGanda += nilai [i][j][0];
        totalNilaiEssay += nilai [i][j][1];
    }
    //Menampilkan hasil
    double pilihanGanda = totalNilaiPilihanGanda / nilai[i].length;
    double essay = totalNilaiEssay / nilai[i].length;
    System.out.printf("Nilai rata-rata ujian soal pilihan ganda siswa " + (i + 1) + " adalah %.2f /n", pilihanGanda);
    System.out.printf("Nilai rata-rata ujian soal essay siswa " + (i + 1) + " adalah %.2f /n", essay);
    System.out.println();
}
}
```

LATIHAN

Latihan 1



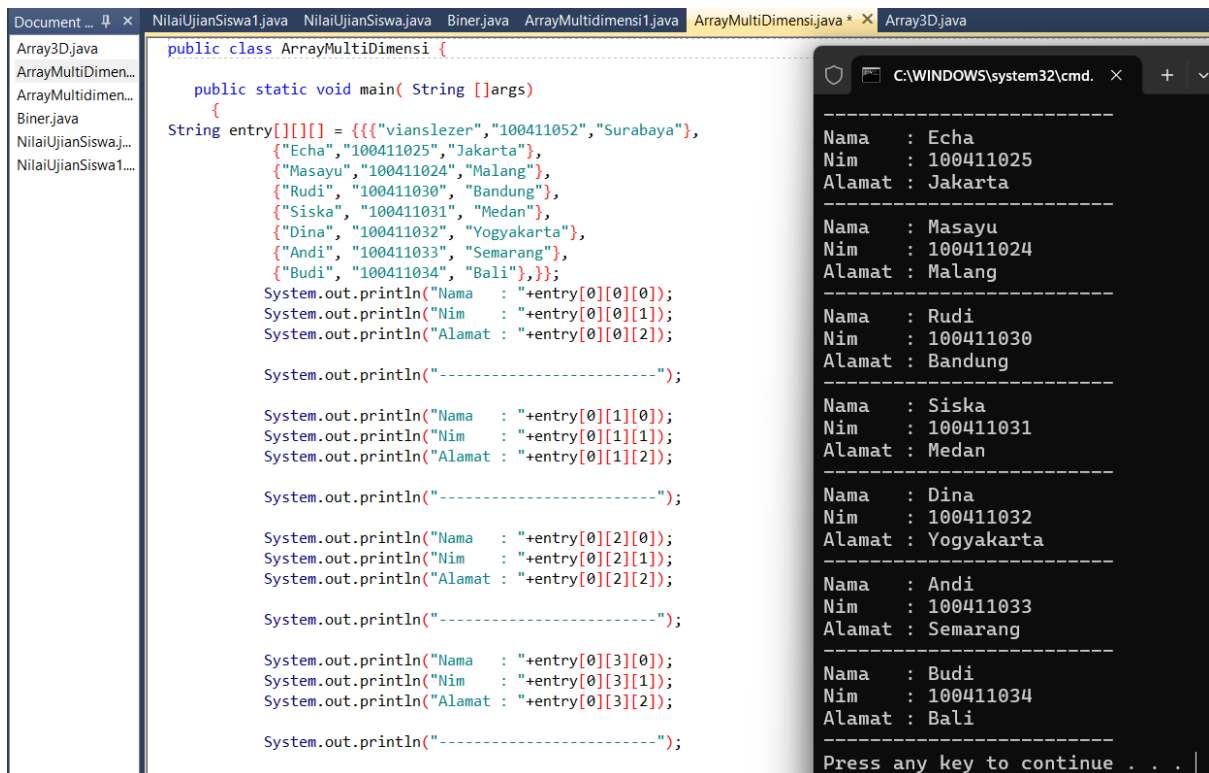
The screenshot shows an IDE with a project named 'NilaiUjianSiswa1'. The file 'ArrayMultidimensi1.java' is open, containing the following code:

```
public class ArrayMultidimensi1 {  
    public static void main( String []args)  
    {  
        String entry[][][] = {{  
            {"vianslezer", "100411052", "Surabaya"},  
            {"Echa", "100411025", "Jakarta"},  
            {"Masayu", "100411024", "Malang"}}};  
  
        System.out.println("Nama : "+entry[0][0][0]);  
        System.out.println("Nim : "+entry[0][0][1]);  
        System.out.println("Alamat : "+entry[0][0][2]);  
  
        System.out.println("-----");  
  
        System.out.println("Nama : "+entry[0][1][0]);  
        System.out.println("Nim : "+entry[0][1][1]);  
        System.out.println("Alamat : "+entry[0][1][2]);  
  
        System.out.println("-----");  
  
        System.out.println("Nama : "+entry[0][2][0]);  
        System.out.println("Nim : "+entry[0][2][1]);  
        System.out.println("Alamat : "+entry[0][2][2]);  
  
        System.out.println("-----");  
    }  
}
```

The command prompt output shows the following results:

```
Nama : vianslezer  
Nim : 100411052  
Alamat : Surabaya  
-----  
Nama : Echa  
Nim : 100411025  
Alamat : Jakarta  
-----  
Nama : Masayu  
Nim : 100411024  
Alamat : Malang  
-----  
Press any key to continue . . . |
```

Latihan 2



The screenshot shows an IDE with a project named 'NilaiUjianSiswa1'. The file 'ArrayMultiDimensi.java' is open, containing the following code:

```
public class ArrayMultiDimensi {  
    public static void main( String []args)  
    {  
        String entry[][][] = {{{"vianslezer", "100411052", "Surabaya"},  
            {"Echa", "100411025", "Jakarta"},  
            {"Masayu", "100411024", "Malang"},  
            {"Rudi", "100411030", "Bandung"},  
            {"Siska", "100411031", "Medan"},  
            {"Dina", "100411032", "Yogyakarta"},  
            {"Andi", "100411033", "Semarang"},  
            {"Budi", "100411034", "Bali"}}};  
  
        System.out.println("Nama : "+entry[0][0][0]);  
        System.out.println("Nim : "+entry[0][0][1]);  
        System.out.println("Alamat : "+entry[0][0][2]);  
  
        System.out.println("-----");  
  
        System.out.println("Nama : "+entry[0][1][0]);  
        System.out.println("Nim : "+entry[0][1][1]);  
        System.out.println("Alamat : "+entry[0][1][2]);  
  
        System.out.println("-----");  
  
        System.out.println("Nama : "+entry[0][2][0]);  
        System.out.println("Nim : "+entry[0][2][1]);  
        System.out.println("Alamat : "+entry[0][2][2]);  
  
        System.out.println("-----");  
  
        System.out.println("Nama : "+entry[0][3][0]);  
        System.out.println("Nim : "+entry[0][3][1]);  
        System.out.println("Alamat : "+entry[0][3][2]);  
  
        System.out.println("-----");  
    }  
}
```

The command prompt output shows the following results:

```
Nama : Echa  
Nim : 100411025  
Alamat : Jakarta  
-----  
Nama : Masayu  
Nim : 100411024  
Alamat : Malang  
-----  
Nama : Rudi  
Nim : 100411030  
Alamat : Bandung  
-----  
Nama : Siska  
Nim : 100411031  
Alamat : Medan  
-----  
Nama : Dina  
Nim : 100411032  
Alamat : Yogyakarta  
-----  
Nama : Andi  
Nim : 100411033  
Alamat : Semarang  
-----  
Nama : Budi  
Nim : 100411034  
Alamat : Bali  
-----  
Press any key to continue . . . |
```

```

System.out.println("Nama    : "+entry[0][4][0]);
System.out.println("Nim     : "+entry[0][4][1]);
System.out.println("Alamat  : "+entry[0][4][2]);

System.out.println("-----");

System.out.println("Nama    : "+entry[0][5][0]);
System.out.println("Nim     : "+entry[0][5][1]);
System.out.println("Alamat  : "+entry[0][5][2]);

System.out.println("-----");

System.out.println("Nama    : "+entry[0][6][0]);
System.out.println("Nim     : "+entry[0][6][1]);
System.out.println("Alamat  : "+entry[0][6][2]);

System.out.println("-----");

System.out.println("Nama    : "+entry[0][7][0]);
System.out.println("Nim     : "+entry[0][7][1]);
System.out.println("Alamat  : "+entry[0][7][2]);

System.out.println("-----");

}
}

```

Latihan 3

The screenshot shows an IDE with a project named 'NilaiUjianSiswa1'. The file 'Biner.java' is open, containing the following code:

```

public class Biner{
    public static void main(String[] args) {
        int[] biner = {0, 1}; // nilai yang mungkin untuk setiap posisi

        // Gunakan array dan perulangan untuk kombinasi
        for (int i = 0; i < biner.length; i++) {
            for (int j = 0; j < biner.length; j++) {
                for (int k = 0; k < biner.length; k++) {
                    System.out.println(biner[i] + " | " + biner[j] + " | " + biner[k]);
                }
            }
        }
    }
}

```

The command prompt output shows the result of the program execution:

```

0 | 0 | 0
0 | 0 | 1
0 | 1 | 0
0 | 1 | 1
1 | 0 | 0
1 | 0 | 1
1 | 1 | 0
1 | 1 | 1
Press any key to continue . . .

```

TUGAS

Tugas 1

-